Madison Montgomery

Final Project Report

For my final data project, I wanted to explore the effect of age on the performance of different positions in professional baseball. Baseball was 'invented' in the 1840's and is closely related to cricket. The first professional team was founded in 1869 with a professional league following soon after. The game is simple, there are two teams competing to score the most runs, gained by running around the bases and crossing home plate, where the player starts. The other team's goal is to stop the runner from scoring, through getting them out in a variety of different ways. The teams switch every 3 outs until each team get 9 turns on each offense and defense. Baseball has changed exponentially over it's history, the balls, bats, and gloves have changed in quality to adjust for the increasing demands of the sport. Solely on the defensive side, rules have changed, throwing speed has increased exponentially, and the number of defensive plays has lowered, mainly due to pitching advancements. While baseball has made continuous leaps forward in performance over the years, baseball is starting to meet its limitation of what the human body can produce. An arm, in a throwing motion, can only exert so much energy before it fails. Longevity of a player has been an important question in the recent years. In the past player could play well into their 40's with no problems, now the stress exerted takes a toll on the players body and a player would be lucky to play into their late 30's. Even then many players face long-term injuries before then, hindering their career. Different positions should, in theory, effect the physical exertion on the body, and therefore effect how long a player could be playing or when their body would start to deteriorate.

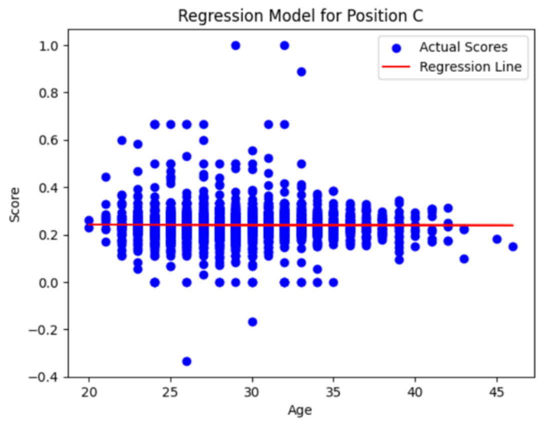
I got the data set I used for my project off of Kaggle. It is titled 'The History of Baseball' and consists of a complete history of major league baseball statistics from 1871 to 2015. Essentially it is a reformatted Lahman's Database, which has a collection of statistics for every team and player in the history of baseball. It contains a total of 24 tables, but for the purposes of this project I only used two of these tables, the fielding table and the player table. The fielding table has 170526 records and18 features, while the player table has 18846 records and 23 features. The fielding table contains fielding statistics like position, errors, games played, etc. all assigned to a player id. The player table contains player information attached to those player id's including name, birthday, and debut date. The first thing done to both tables was getting rid of unnecessary columns such as player name, before joining the two tables by the player id column. This assigned each record with the personal information of that player. I then did a second round of getting rid of unnecessary columns like height and weight. I then took the year and subtracted each records birthyear to record the player age during that season to the newly created age column. I then removed all records where the season was older than 1985 to account for baseball advancements, the 1980's could be seen as the start of a 'modern' era of baseball. This left me with 38741 records to use for analysis. I then removed all columns that could infer the year of the record, including year and leagues. This left me with a 38741 rows x 14 columns table by which I could begin to analyze.

With a cleaned table ready to be analyzed, I began by categorizing the table by position. I lumped together left, right, and center field into outfield, ‘of’, due to them being so similar that some records have the player playing more than one. I then separated the catchers ‘c’, out of the table and into a table called c_data. Catchers have stats that don’t apply to the rest of the positions so it was easier to separate it and delete those columns for the table containing the rest of the positions. I then added a ‘score’ column to both tables and calculated a score that would assess performance. Still to this day there are arguments over which statistics matter more in player evaluation, this becomes even more uncertain when regarding defensive statistics. With this in mind I calculated score based on the base idea of performance, runs created for offense and runs saved for defense. With this in mind, regular position’s score was created by taking their putouts, the number of outs they physically made, adding assists, the number of plays he assisted in making, and subtracting errors, the number of mistakes that allowed an out to not be made. This would give a picture of how many he runs he saved through outs. I divided all this by inn_outs, which is the number of innings played quantified by outs, for example 1 inning is 3 outs. This accounts for playing time. For catchers I still took putout and added them to assists, but instead of subtracting errors, I subtracted the number of bases the opposing team stole and the number of pass balls, effectively punishing the giving up of extra bases. Then I still divided by inn_outs to account for playing time. With everything calculated, I separated all positions into their own individual graphs, making it easier to evaluate based upon positions. I first got the summary statistics of each graph, which include their count, mean, standard deviation, etc.

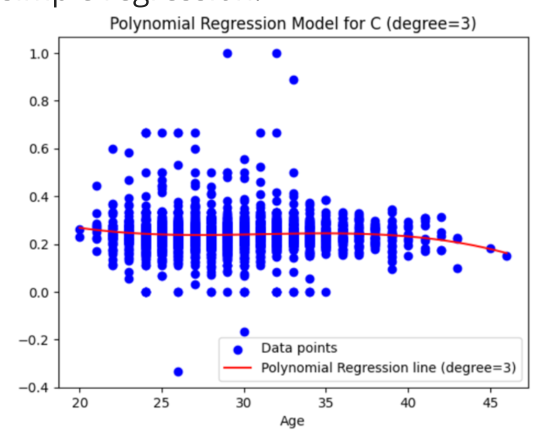
```
summary_stats = {}

# Compute summary statistics for each position
for position, df in {'C': c_data, 'P': p_data, '1B': b1_data, '2B': b2_data, '3B': b3_data, 'SS': ss_data, 'OF': of_data}.items():
    summary_stats[position] = df['score'].describe()

# Create a DataFrame from the summary statistics
summary_df = pd.DataFrame(summary_stats)

# Transpose the DataFrame for better readability
summary_df = summary_df.T

# Display the summary statistics
print("Summary Statistics for Each Position:")
print(summary_df)
```

```
Summary Statistics for Each Position:
      count      mean       std       min       25%       50%       75%  \
C    2460.0  0.240032  0.070923 -0.333333  0.208969  0.237190  0.266667
P   18095.0  0.059498  0.042496 -0.500000  0.037612  0.056911  0.076923
1B   5032.0  0.342166  0.114090 -0.333333  0.315667  0.340692  0.371475
2B   3377.0  0.176780  0.070016 -0.333333  0.160000  0.179253  0.198693
3B   1306.0  0.087599  0.055265 -0.500000  0.077700  0.092016  0.103133
SS    980.0  0.160863  0.039290 -0.111111  0.150306  0.161926  0.173846
OF   5732.0  0.083827  0.041453 -0.333333  0.069037  0.082841  0.097102

         max
C   1.000000
P   1.000000
1B  3.333333
2B  1.000000
3B  0.555556
SS  0.666667
OF  1.000000
```

I then created basic regression models with lines of best fit for each of the positions individually.

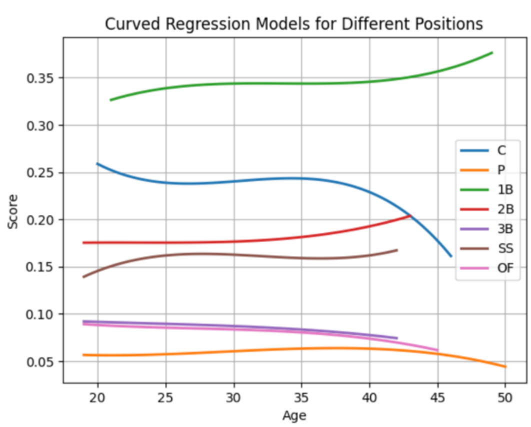


After this I created training and test sets to create a basic regression graph with a curved regression line for each graph in order to show a more accurate picture of the simple regression.



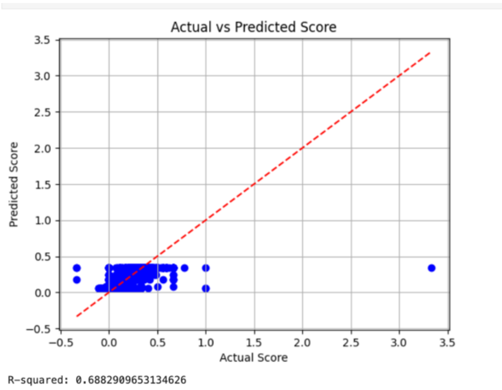
.

I then layered and color-coded the models over each other to compare their regression lines with each other.

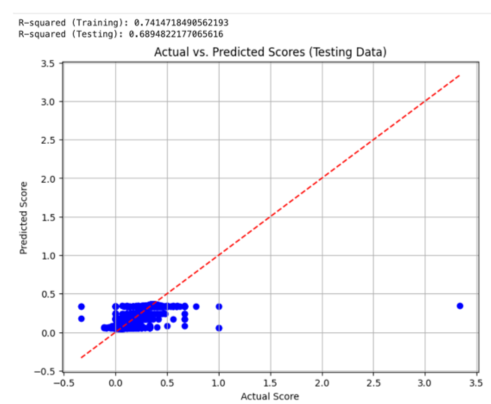


Essentially this overlayed model allows the comparison of peak performance ages for each of the positions to each other. From this model we can see that positions typically improve from the time they start but plateau in their mid 20’s. the exception being catchers who see an immediate dip in performance in their early 20s before they plateau. Additionally about half of positions have their performance start a downward curve once they reach the age of 35. The exceptions being short stop and second base who's performance sees a slight upturn. The most significant drop-off in performance is seen in catchers who go from performing at the .25 level to almost a .15 level within ten years. Essentially this model allows us to observe trends in different positions performances over time.

Using the data collected I created a multiple regression model of the data.



I used age and position as the input values to calculate the score, using the test and train approach. The model compares the predicted scores with the actual scores. The R-squared of my model calculated to .69, meaning that approximately 70% of the variance in score can be explained by the age and position of the player. This means that the age of a player appears to have a significant impact on player performance. Also the positions influence the score, there may be distinct differences in the trends of these positions due to responsibilities and playing styles. While this model performs reasonably well it is limited by factors not represented in the input. I explored creating the multiple regression model with more inputs, such as height, weight, double plays, games, throws, and bats.



The highest scoring of these multiple regression models was the model with all of these variables as input, which has a .6894 r^2 score, the original, with just age and position, had a fractionally smaller r^2 score with a .6882. This indicates that out of all these variables, age and position have the most effect on player performance. This indicates that the other factors that influence player performance cannot be found in either the player or fielding data. For example player development and experience could influence performance at any age as well as factors like team strategies and locations. Additionally I decided to do a ten-fold cross-validation on the first, the age and position input, multiple regression model.

```
Model Coefficients:
[ 0.17636603  0.01258472 -0.07750667  0.07651909 -0.08024597 -0.10487681
 -0.00284039  0.00023343]
Intercept: 0.15764239107702585
Mean Squared Error: 0.00450618246162656
Ten-Fold Cross-Validation Scores: [       nan 0.00169435 0.00181615 0.00193438 0.00151659 0.00726391
 0.01075174 0.00470734        nan 0.00125572]
```

For this dataset the mean squared error calculates to 0.0045 which is relatively low and indicates the models predictions are generally close to the actual value. The ten-fold cross-validation scores for this data set range from approximately 0.0013 to 0.0108, with the average being 0.0045. these numbers would seem to indicate that the model is performing generally well, however for two of the folds the score is NaN. This indicates that there may issues with either the model or the data. In an effort to remedy the NaN's I investigated taking out all null values and outliers, however the output remained the same. This may indicate the model is over-fit or there is a logic error in my code. To enhance this model it could have additional training with further factors that could effect score and exploring different modeling techniques. It could also be improved by a deep investigation as to the root cause of the NaNs in the cross-validation, or explorations into other cross-validation techniques that may perform better. In summary, while my model shows a correlation between and player performance at different positions, it has room for improvement. The insight from this model could be used to help teams make decisions about player contracts, in terms of pay and contract length. It could also help teams predict future performance for players and give a guide of where a player of a certain position and age should be performing.

Citations

https://www.kaggle.com/datasets/seanlahman/the-history-of-baseball?resource=download
zyBooks, a Wiley brand. http://www.zybooks.com (accessed 2024).